

# Cosine Similarity

Cosine similarity compares the orientation of two vectors rather than their absolute magnitude and is widely used for semantic retrieval.

## Overview

Cosine Similarity is a practical part of a document question-answering pipeline. Its purpose is to make information easier to process, retrieve, validate, and use reliably. A well-designed implementation should expose clear inputs, outputs, metadata, and failure behavior.

## Key Engineering Principles

1. Keep this stage modular so it can be tested independently.
2. Preserve document and source metadata whenever information moves between pipeline stages.
3. Evaluate changes with representative technical questions rather than a single demonstration query.
4. Prefer controlled failure behavior over silently producing unsupported results.

## Practical Example

Consider a user asking a technical question related to Cosine Similarity. The application processes the query, identifies useful evidence, and passes that evidence to downstream components. The final response should remain within the available evidence and expose enough source information for verification.

## Implementation Notes

When implementing Cosine Similarity, engineers should consider correctness, latency, resource usage, maintainability, and security. One useful development method is to change one major variable at a time and rerun a fixed evaluation set so that improvements can be measured consistently.

## Summary

The central idea is to treat Cosine Similarity as a measurable component of the overall RAG pipeline. Reliable inputs, explicit configuration, source traceability, controlled errors, and repeatable evaluation make the resulting system easier to improve.

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